

Access-profile detail

Actual acquisition states are distinct from the historical M2 accessibility scores in Figures 2–4.

Task		Content access		Official content detail		Source version clarity	
		CANN	CUDA‡	CANN	CUDA‡	CANN	CUDA‡
Environment and installation							
I	Version compatibility	C	C	4	5	4	4
J	Installation	C	C	5	5	2	3
K	Container setup	C	C	5	4	4	5
Operator development							
C	Library selection	P	C	3	5	3	5
D	Custom operator	N	C	—	5	2	3
L	Operator accuracy	C	C	5	5	2	5
M	Dynamic shapes / tiling	C	C	4	5	4	4
N	Operator fusion	C	C	5	4	2	5
O	Framework integration	C	C	5	5	3	3
Training							
F	Distributed training	C	C	4	5	2	3
P	Mixed precision	C	C	4	5	4	5
Q	Memory / OOM	C	C	5	5	4	5
R	Training convergence	C	C	5	4	3	5
Inference and deployment							
A	Model conversion	C	C	4	5	2	4
H	Quantization	C	C	3	5	2	3
S	Inference serving	C	C	4	5	3	4
T	Dynamic inference	C	C	5	5	2	5
Performance and optimization							
B	Profiling	C	C	4	5	2	4
U	Memory / occupancy	C	C	4	5	5	5
V	Compute-transfer overlap	C	C	4	5	2	5
W	Multi-device communication	C	C	3	4	3	4
Debugging							
E	Error-code diagnosis	C	C	2	5	2	5
X	Memory-error diagnosis	C	C	5	5	2	3
Migration							
Y	Chip migration	P	C	4	5	3	4
Z	Programming concepts	C	C	5	4	5	5
Migration analogy (not in CANN/CUDA summaries)							
G	Migration analogy [CANN / ROCm-HIP]	C	C	4	5	4	3

Content status: C = core obtained; P = partial content obtained; N = not obtained. Access route is recorded separately in the protocol.

‡ For G, CUDA‡ = ROCm/HIP; it is a migration analogy and is excluded from CANN/CUDA summaries.